

ACCTG 211H — Mechanics Drill Set (Answer Key)

Week 10 — Equity, EPS, and Stock-Based Compensation

Drill Set

Quick, single-skill reps to build mechanical fluency before the week's first class. Graded for **completion** — work each one by hand, then check it against the key that follows and re-do any you missed.

1. (*Issue stock above par.*) A firm issues 10,000 shares of \$1 par common stock for \$15 per share. Record the journal entry.
2. (*Treasury stock, cost method.*) The firm repurchases 1,000 of its own shares for \$18 each. Record the journal entry.
3. (*Cash dividend.*) A firm declares a \$0.50 per share cash dividend on 200,000 shares outstanding. Record (a) the declaration and (b) the payment.
4. (*Basic EPS.*) Net income is \$2,400,000, preferred dividends are \$200,000, and the weighted-average number of common shares outstanding is 1,000,000. Compute basic earnings per share.
5. (*Weighted-average shares.*) A firm has 800,000 shares outstanding on January 1 and issues 240,000 additional shares on July 1. Compute the weighted-average shares outstanding for the year.
6. (*Diluted EPS, treasury-stock method.*) A firm has net income of \$1,800,000 and 1,000,000 weighted-average shares. It has 100,000 options outstanding with a \$20 exercise price; the average market price is \$25. Compute diluted EPS.
7. (*Stock-based compensation expense.*) A firm grants options with a total grant-date fair value of \$600,000 that vest evenly over a three-year service period. Compute the annual compensation expense and record one year's entry.
8. (*RSU expense, partial year.*) A firm grants 12,000 restricted stock units on July 1 with a grant-date fair value of \$30 each, vesting evenly over four years. Compute the compensation expense for the first calendar year (July–December).

Self-Check Answers

1. **Issue above par.** Dr Cash \$150,000 ($10,000 \times 15$); Cr Common Stock \$10,000 ($10,000 \times 1$ par); Cr Additional Paid-in Capital \$140,000.
2. **Treasury stock.** Dr Treasury Stock \$18,000 ($1,000 \times 18$); Cr Cash \$18,000.
3. **Cash dividend.** (a) Declaration: Dr Retained Earnings (Dividends) \$100,000 ($200,000 \times 0.50$); Cr Dividends Payable \$100,000. (b) Payment: Dr Dividends Payable \$100,000; Cr Cash \$100,000.
4. **Basic EPS.** = (Net income – preferred dividends) $\div$ weighted-average shares = $(2,400,000 - 200,000) \div 1,000,000 = \mathbf{\$2.20}$.
5. **Weighted-average shares.** = $800,000 + 240,000 \times (6/12) = 800,000 + 120,000 = \mathbf{920,000}$ shares.
6. **Diluted EPS.** Proceeds from exercise = $100,000 \times \$20 = \$2,000,000$. Shares repurchased at average price = $2,000,000 \div 25 = 80,000$. Incremental (net new) shares = $100,000 - 80,000 = 20,000$. Diluted shares = $1,000,000 + 20,000 = 1,020,000$. Diluted EPS = $1,800,000 \div 1,020,000 = \mathbf{\$1.76}$ (down from basic EPS of \$1.80).
7. **Option expense.** Annual = $600,000 \div 3 = \mathbf{\$200,000}$. Entry each year: Dr Compensation Expense \$200,000; Cr Additional Paid-in Capital — Stock Options \$200,000.
8. **RSU expense.** Total grant fair value = $12,000 \times \$30 = \$360,000$. Annual expense = $360,000 \div 4 = \$90,000$. First calendar year is a half-year (July–December): $90,000 \times 6/12 = \mathbf{\$45,000}$.